

CHP-2 Electric Field

General Physics 2

2026-09-02

Electric Field

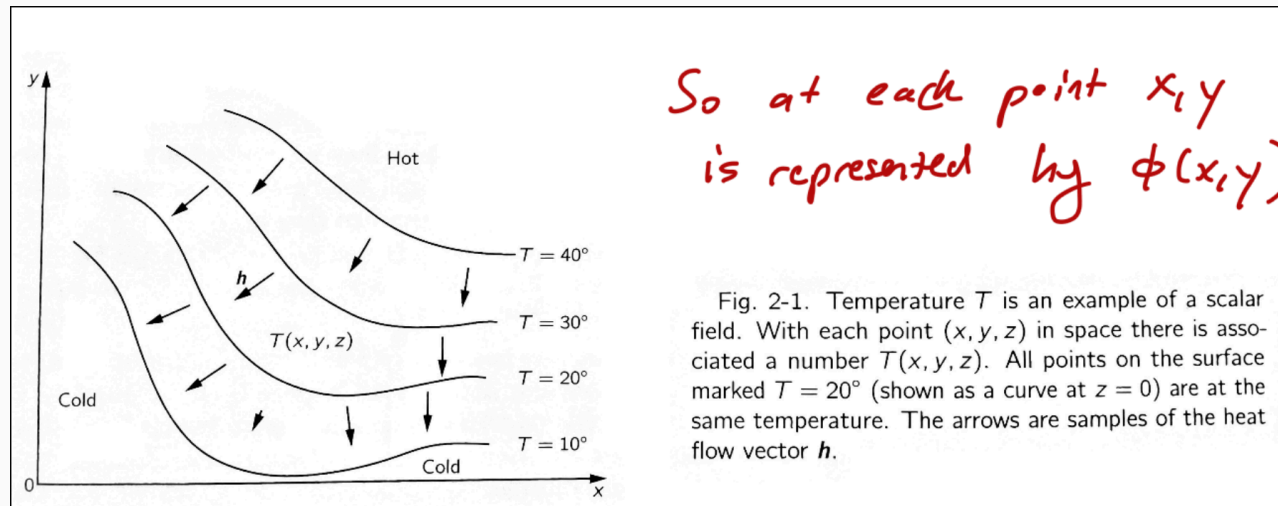
Fields

Up to this point (including previous semester) we have taken the view that gravitational and electric forces between particles are action-at-a-distance, that is a particle exerts a direct gravitational or electric force on another particle even if they are not touching. This is what bothered Newton alot, about which he says:

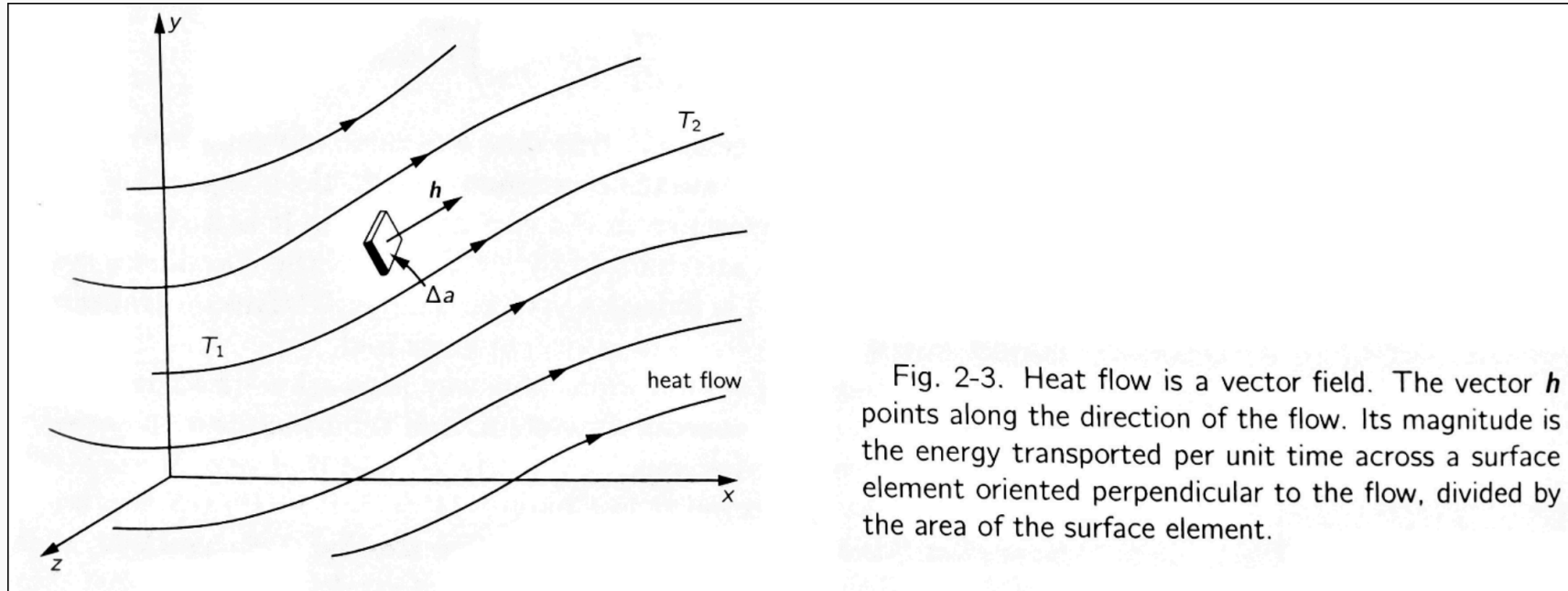
// It is inconceivable, that inanimate brute matter, should, without the mediation of something else, which is not material, operate upon and affect other matter without mutual contact. That Gravity should be innate, inherent and essential to Matter so that one Body may act upon another at a Distance thro' a *Vacuum* without the Mediation of anything else, by and through which their Action and Force may be conveyed from one to another, is to me so great an Absurdity that I believe no Man who has in philosophical Matters a competent Faculty of thinking can ever fall into it. Gravity must be caused by an Agent acting constantly according to certain Laws; but whether this Agent be material or immaterial, I have left to the consideration of my readers. //

Field: In order to explain this action-at-a-distance in modern view we will introduce field as the mediator of the forces. So a gravitating or electrically charged body generates a gravitational or electric field.

Scalar field: The simplest field is the scalar field. By scalar field we mean a field which is characterised at each point by a single number or a scalar. So at each point x, y is represented by $f(x, y)$.



Vector field: For the vector fields idea is same but this time there are vectors given for each point in space.



Electric Field

- So far we have worked Coulomb's law in terms of the force between charges. However, this understanding leads to serious difficulties when charges start to move.
- If we move a charge q' somewhat near to charge q then the electric force must increase. But increase can not occur instantly! (No signal can propagate faster than light). So when we move charge q' some kind of disturbance travels through empty space from q to q' at the speed of light. This charges exert forces on one another by means of disturbances that they generate in the space surrounding them. These disturbances are electric fields.
- The fields are endowed with energy and momentum.

The formal definition of electric field is as follows: To discover the field at given position, take a point charge q' (“test charge”) place it in position. The charge will feel an electric force $\vec{F}$, the electric field $\vec{E}$ at that point is defined by.

$$\vec{E} = \frac{\vec{F}}{q'}$$

The electric field generated by point charge q at distance r is then

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Note that $\vec{E}$ is a vector!

Table 23.1 SOME ELECTRIC FIELDS

At surface of uranium nucleus	$2 \times 10^{21} \text{ N/C}$
At surface of pulsar	$\sim 10^{14} \text{ N/C}$
At orbit of electron in hydrogen atom	$6 \times 10^{11} \text{ N/C}$
In X-ray tube	$5 \times 10^6 \text{ N/C}$
Electrical breakdown of air	$3 \times 10^6 \text{ N/C}$
In Van de Graaff accelerator	$2 \times 10^6 \text{ N/C}$
Within lightning bolt	10^4 N/C
Under thundercloud	$1 \times 10^4 \text{ N/C}$
Near radar transmitter (FPS-6)	$7 \times 10^3 \text{ N/C}$
In sunlight (rms)	$1 \times 10^3 \text{ N/C}$
In atmosphere (fair weather)	$1 \times 10^2 \text{ N/C}$
In beam of small laser (rms)	$1 \times 10^2 \text{ N/C}$
In fluorescent lighting tube	10 N/C
In radio wave	$\sim 10^{-1} \text{ N/C}$
Within household wiring	$\sim 3 \times 10^{-2} \text{ N/C}$
In thermal radiation in intergalactic space (rms)	$3 \times 10^{-6} \text{ N/C}$

Lines of Electric Field

There are different ways to imagine electric fields. One way is to represent graphically by field lines.

(1) Field lines are drawn in such a way that, at any given point, the tangent to the line has the direction of the electric field. The density of lines is directly proportional to the magnitude of the electric field.

(2) We must begin each line on a positive point charge and end on a negative point charge. The positive charges are then sources and negative charges are sinks.

Note that although it is useful to think of electric fields in terms of lines it is not reliable. One has to be careful when the charges are moving! Everything will look different and it's hard to imagine what will be the result. So it might be best to think of field as an abstract idea

Fig. 23.9 (top left) Electric field vectors surrounding a positive point charge. The magnitudes of these field vectors decrease in inverse proportion to the squares of their distances from the point charge (the distances are reckoned from the tails of the vectors to the point charge).

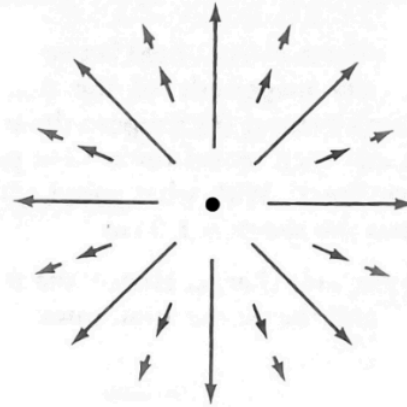
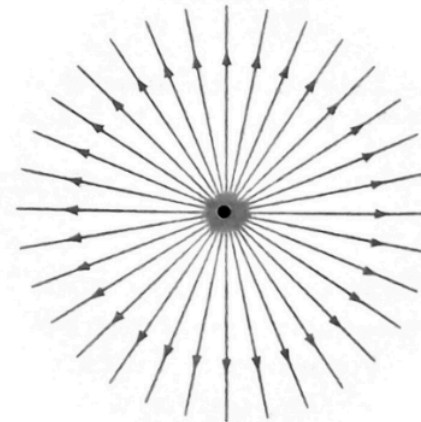
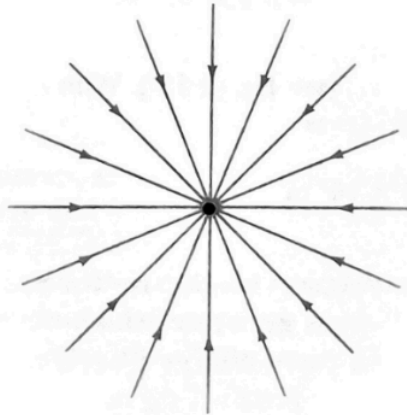
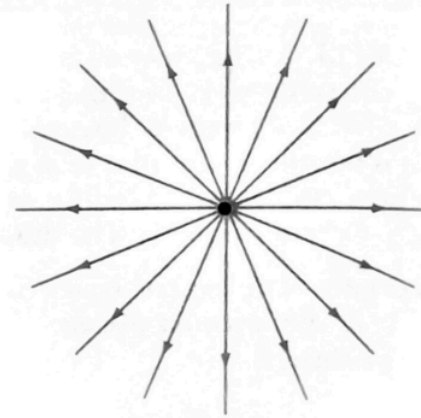


Fig. 23.10 (top right) Electric field lines of a positive point charge. Note that in three dimensions the lines spread out in all three directions of space, whereas the diagram shows the lines spreading out only in the two dimensions within the page. This gives a misleading impression of the density of field lines as a function of distance. This limitation of two-dimensional diagrams should always be kept in mind when looking at pictures of field lines.



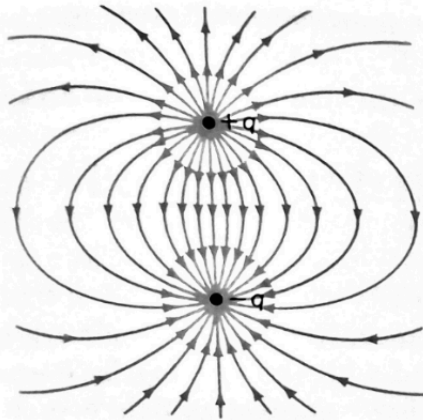


Fig. 23.13 (top left) Field lines generated by positive and negative charges of equal magnitudes.

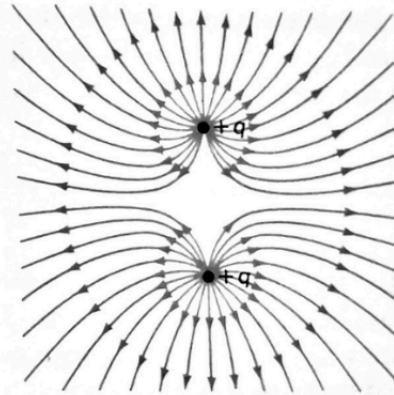


Fig. 23.14 (top right) Field lines generated by two positive charges of equal magnitudes.

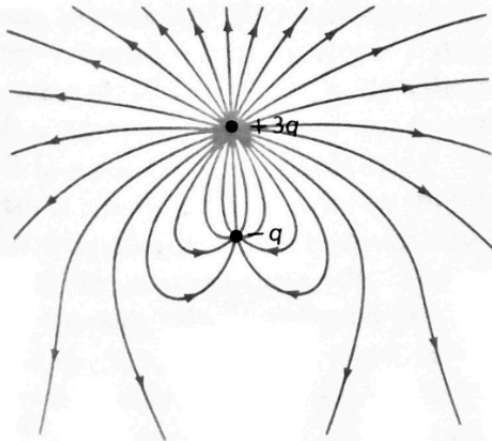


Fig. 23.15 (bottom left) Field lines generated by positive and negative charges of unequal magnitudes. The positive charge has three times the magnitude of the negative charge.

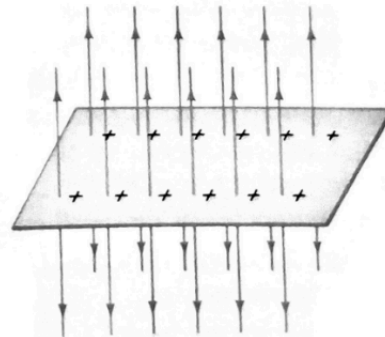


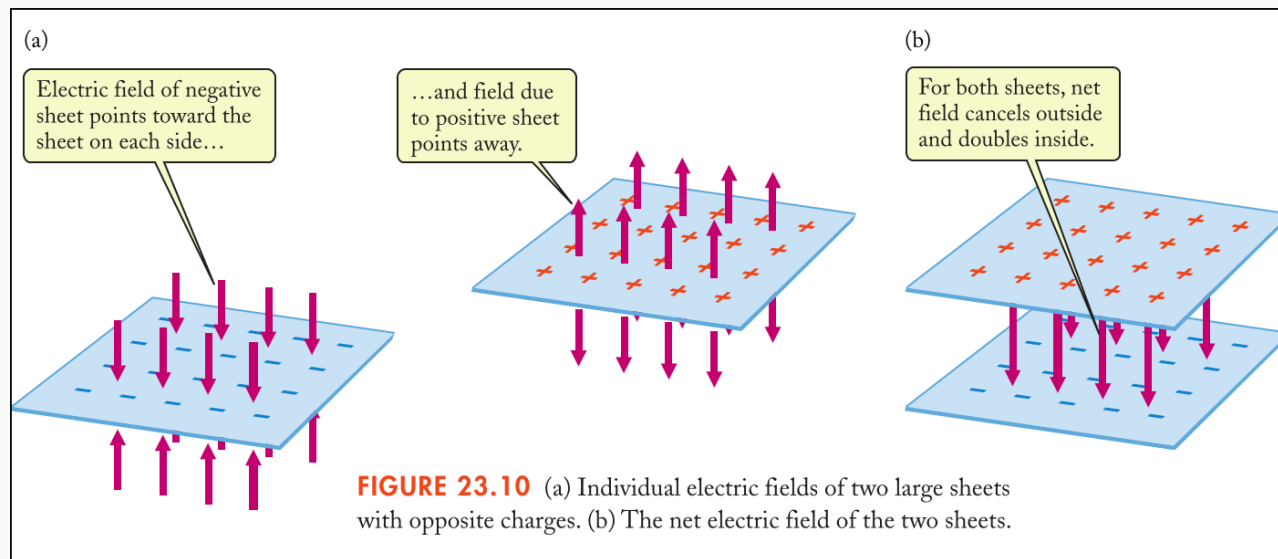
Fig. 23.16 (bottom right) Field lines of a very large sheet of charge.

Example: A total amount of charge Q is distributed uniformly along the circumference of a thin glass ring of radius R . What is the electric field on the axis of the ring?

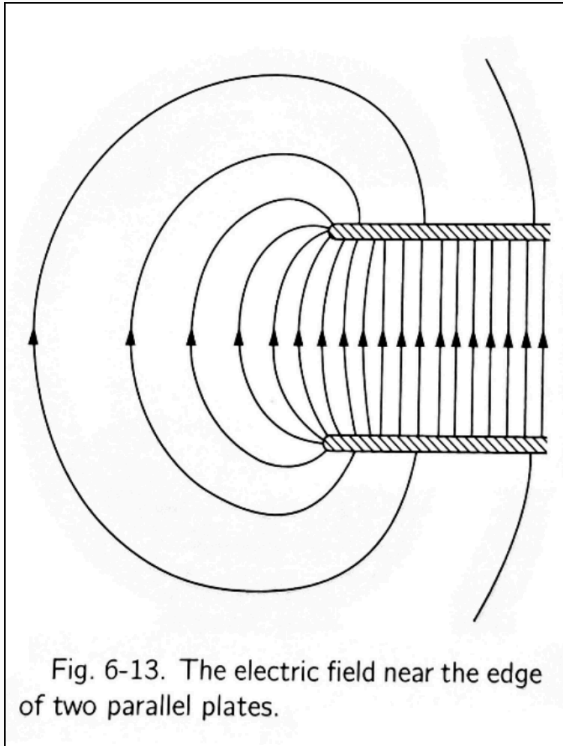
Example: Using above result find the electric field generated by a large flat sheet (like paper) carrying a uniform charge density σ ?

Using the result of the above example and principle of superposition it can be shown that a large charged plate has uniform electric field with a magnitude $E = \sigma/2\epsilon_0$. In practice, two parallel charged sheets are used and together they give a net field.

$$E = \frac{\sigma}{\epsilon_0}$$



- The above figure discussion is for “infinite plates” where in real life with finite plates the field outside does not totally cancel and we have fringes. It is easier to demonstrate in two dimensions.



- Note that near the plate ends the electric field is not homogenous, which will cause the fringes at the end of the plates.

Electric Dipoles

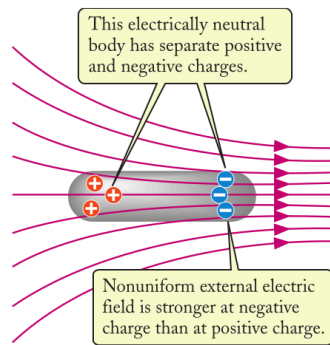


FIGURE 23.24 An electric dipole placed in a nonuniform electric field.

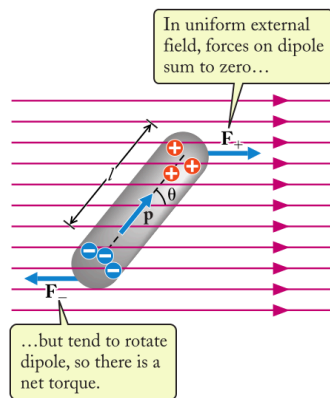


FIGURE 23.25 Electric dipole placed in a uniform electric field. The dipole moment vector $\mathbf{p}$ is directed from the negative charge to the positive charge.

- One might expect that if an electrically neutral body is placed in a given electric field, the body will experience no force.
- Consider the object in the left figure that has positive charges in one end and negative at other. Total charges are zero therefore net force is zero $\vec{F}_{\text{tot}} = 0$.

However, there is a net non-zero torque on system.

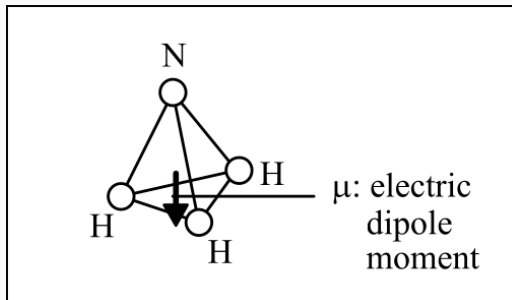
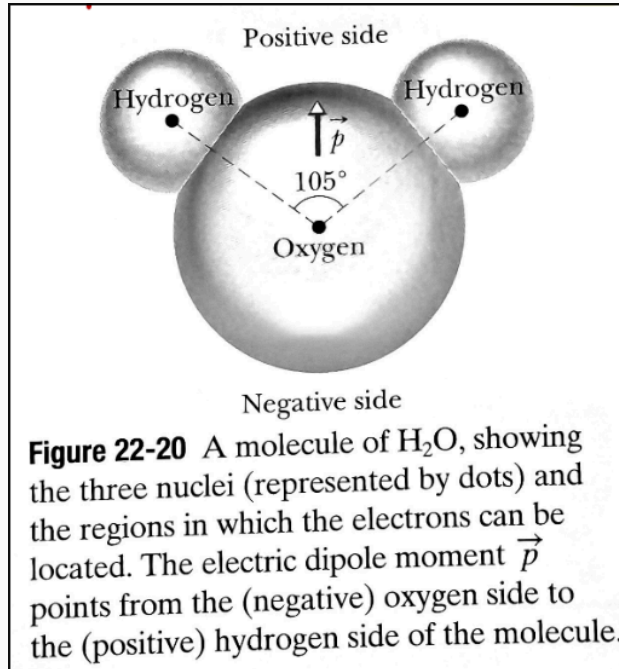
$$\tau = -pE \sin(\theta)$$

- One can define $p = lQ$ as **dipole moment** which is simply the charge multiplied by the separation between the charges. And direction of the dipole moment $\vec{p}$ is from negative to positive charge, then we can write the torque

$$\vec{\tau} = \vec{p} \times \vec{E}$$

- Corresponding to this torque there exists a potential energy that equals to amount of work you must do against the electric forces to twist the dipole through some angle.

$$U = -\vec{p} \cdot \vec{E}$$



- In a water molecule strictly 10 electrons of the molecule tend to remain closer to the oxygen nucleus than the hydrogen which makes the oxygen side more negative.

This **permanent dipole** property of water is what we use in microwave ovens to cook our meal.

- Not all molecules have permanent dipole moments. But they may acquire temporary dipole moment when placed in an electric field. These are called **induced dipole moment**.

The electric field line of the electric dipole moment [$q = 1$ at the point $(0, 1)$ and $q = -1$ at the point $(0, -1)$] is schematically shown as follows.

